

IIS2MDC Offset Cancellation Investigation Report

Bench measurement — 21 July 2026

Results

The IIS2MDC magnetometer has a built-in feature, set by one bit in its configuration register (the OFF_CANC bit in CFG_REG_B), that removes the sensor's own intrinsic magnetic offset. This report compares the sensor's output with that feature turned on and turned off.

Two things appear in the reading when offset cancellation is turned off that are not there when it is on:

- A fixed offset of about 301 mG (roughly 62% of the Earth's magnetic field) is added to the reading.
- That offset changes with temperature, by about 9 mG for every degree Celsius.

With offset cancellation on, neither is present: the reading stays centered and stable. The amount of noise in the reading is similar whether the feature is on or off.

What we tested

Offset cancellation works by taking two readings — one after a magnetic 'set' pulse and one after a 'reset' pulse — and averaging them, which cancels the sensor's intrinsic offset (ST application note AN5080, section 8). It is switched on or off by a single configuration bit.

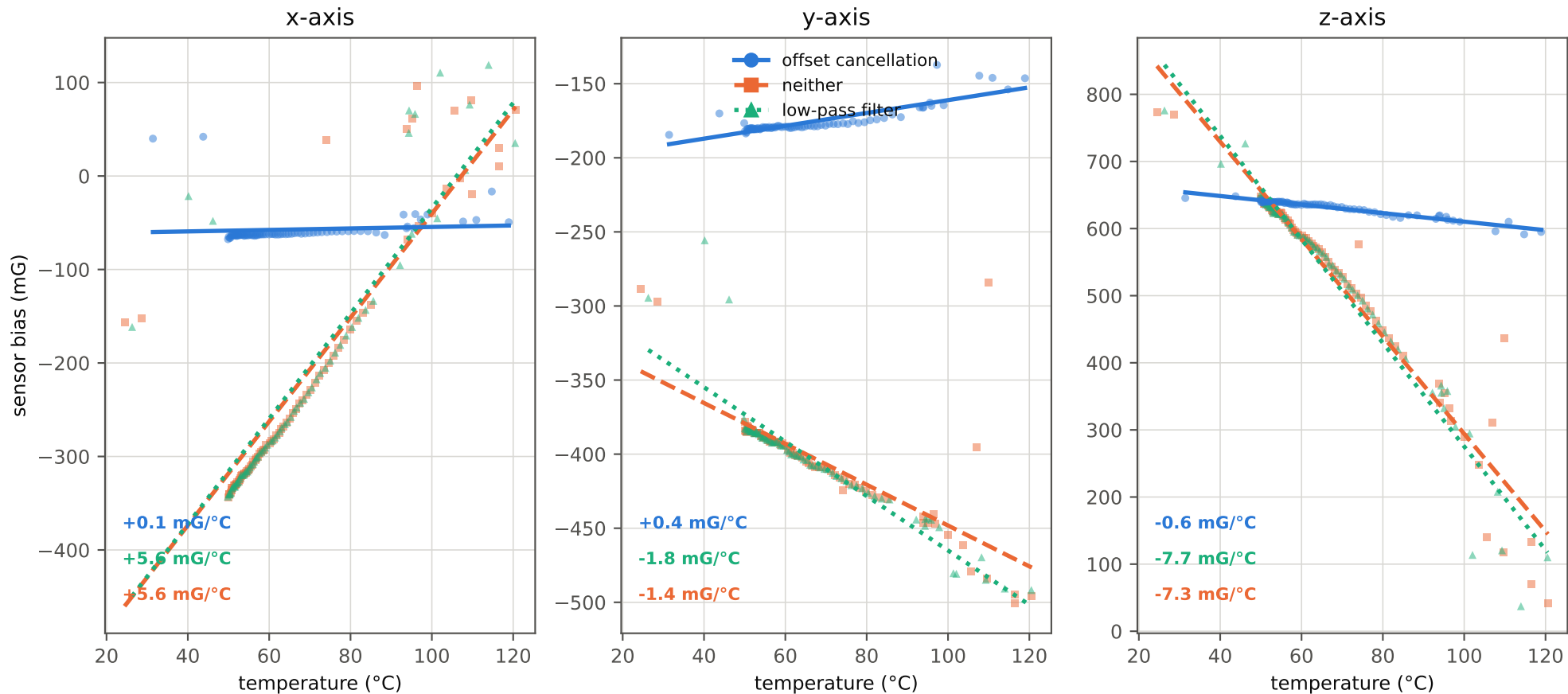
To compare fairly, we switched the sensor between three settings every two seconds while recording its output: offset cancellation, a low-pass filter only, and neither. Because all three settings are exercised within the same recording, the surrounding magnetic field, temperature, orientation and position are the same for each — so any difference between them comes only from the setting.

The low-pass-filter setting is a reference. According to the datasheet it has the same bandwidth and smoothing as offset cancellation but does not remove the offset. So an effect that appears with offset cancellation but not with the low-pass filter comes from the offset removal itself, not from the smoothing. Temperature is read from a separate sensor on the same board, because the magnetometer's own temperature output also changes with the setting.

We recorded three situations: the sensor rotated through all orientations, warmed up and then left to cool, and mounted on a vehicle in flight with the motors running. The plots that follow present the data from these recordings.

Sensor bias versus temperature

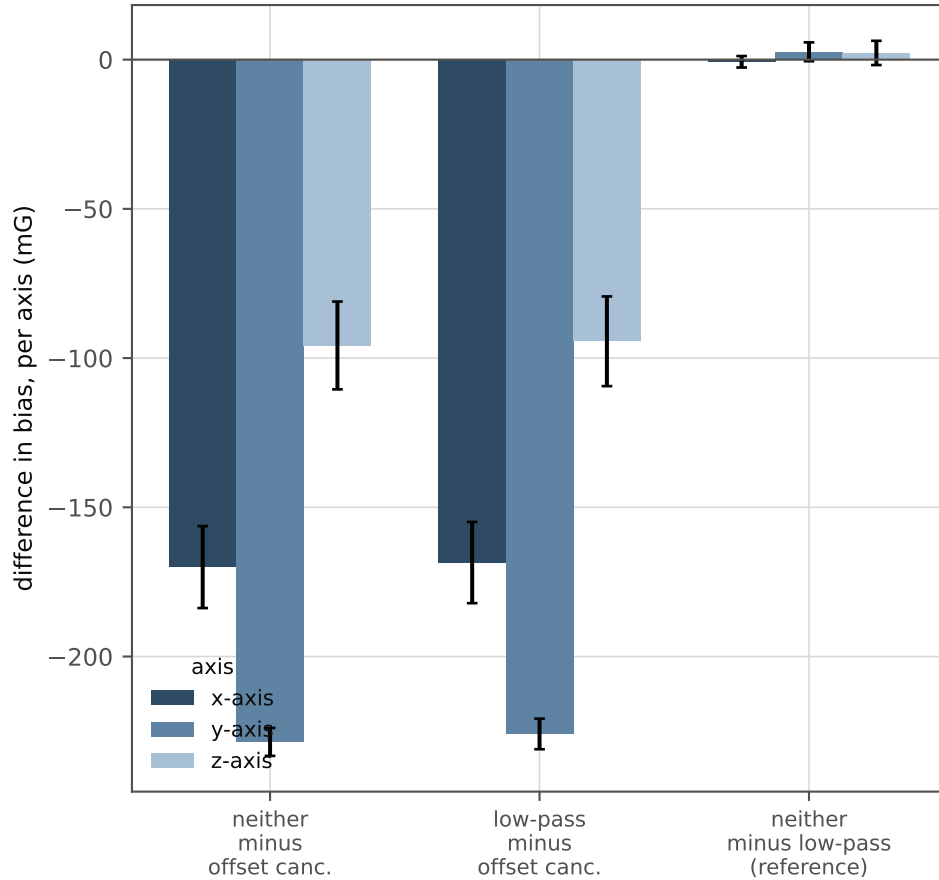
Each point is the sensor's bias during one still two-second window. Temperature is from a separate sensor on the same board.



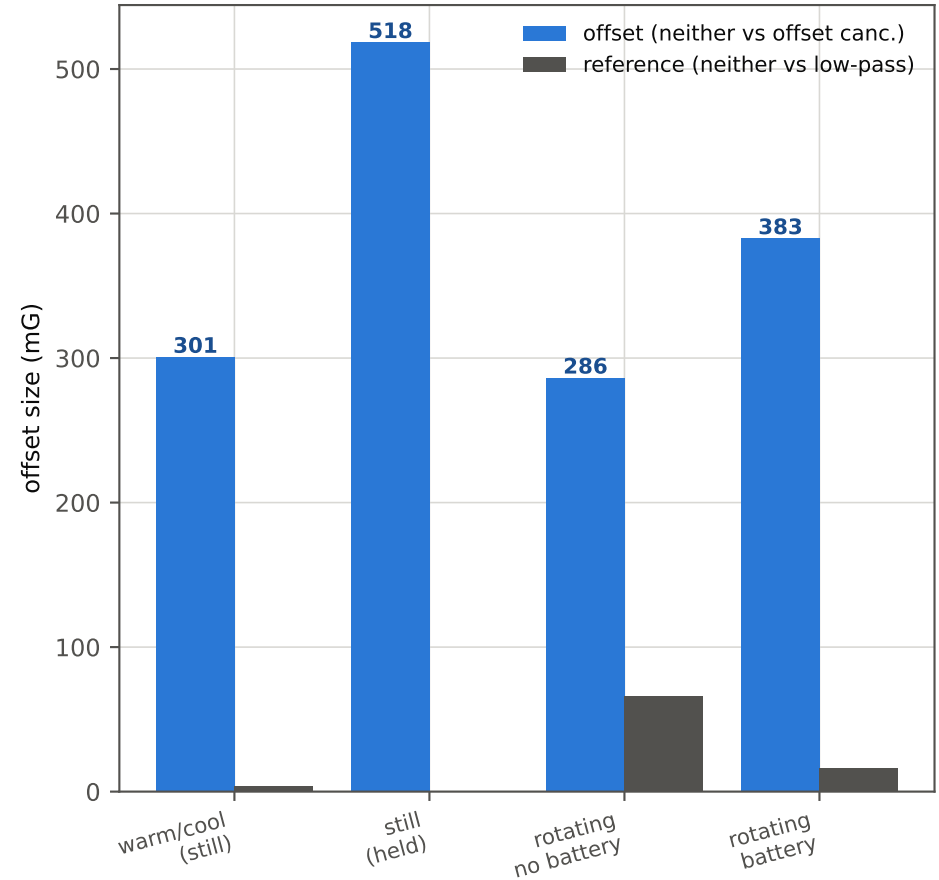
With offset cancellation off (orange) or the low-pass filter only (green), the bias moves steadily with temperature— about 9 mG per °C. With offset cancellation on (blue) the bias barely changes. The blue line stays flat across the same temperature range the others sweep, so the drift comes from the sensor itself and not from the surroundings. Over a 30 °C change that is about 270 mG of bias shift.

Fixed offset added when offset cancellation is off

Difference between settings, held still

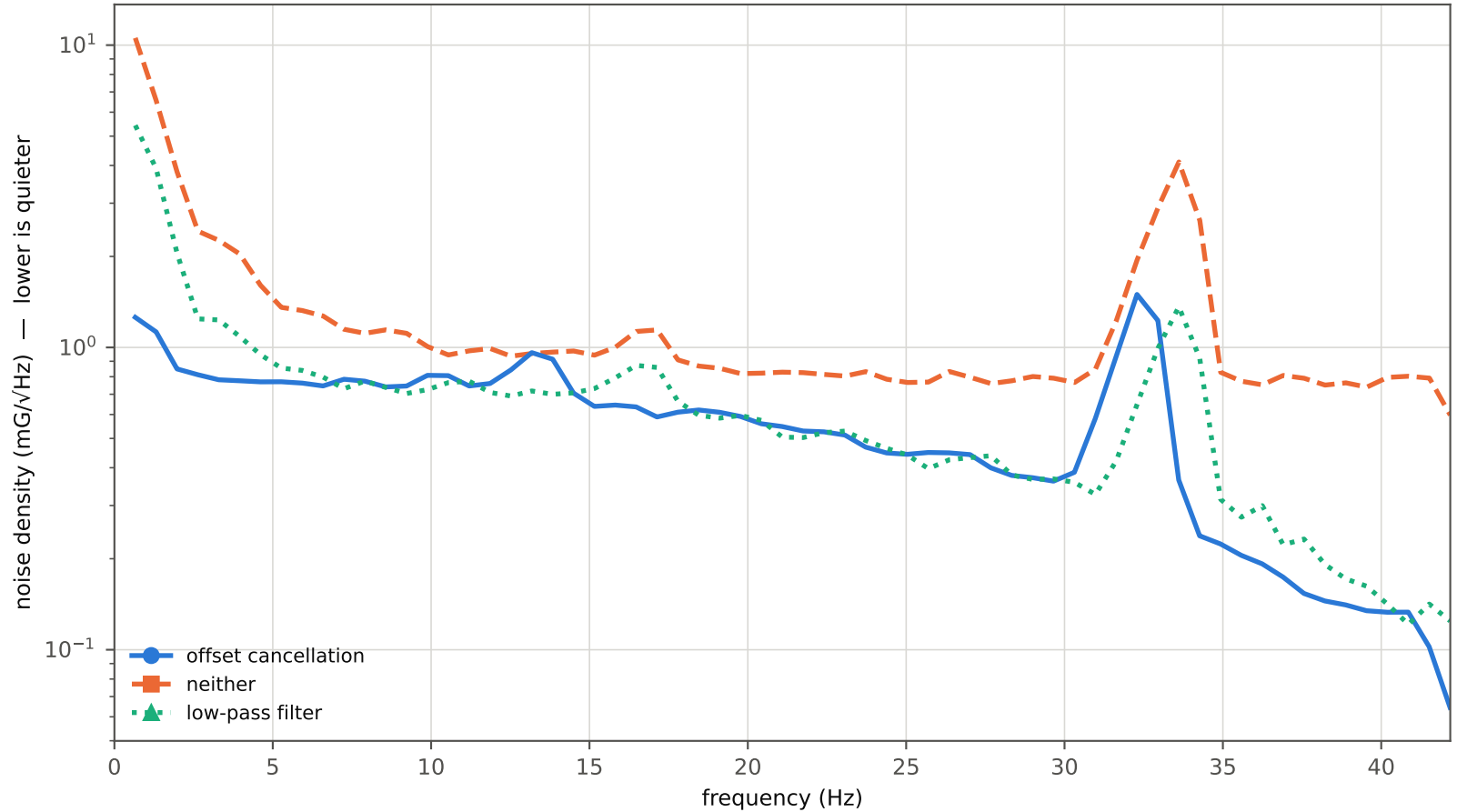


Offset size in each recording



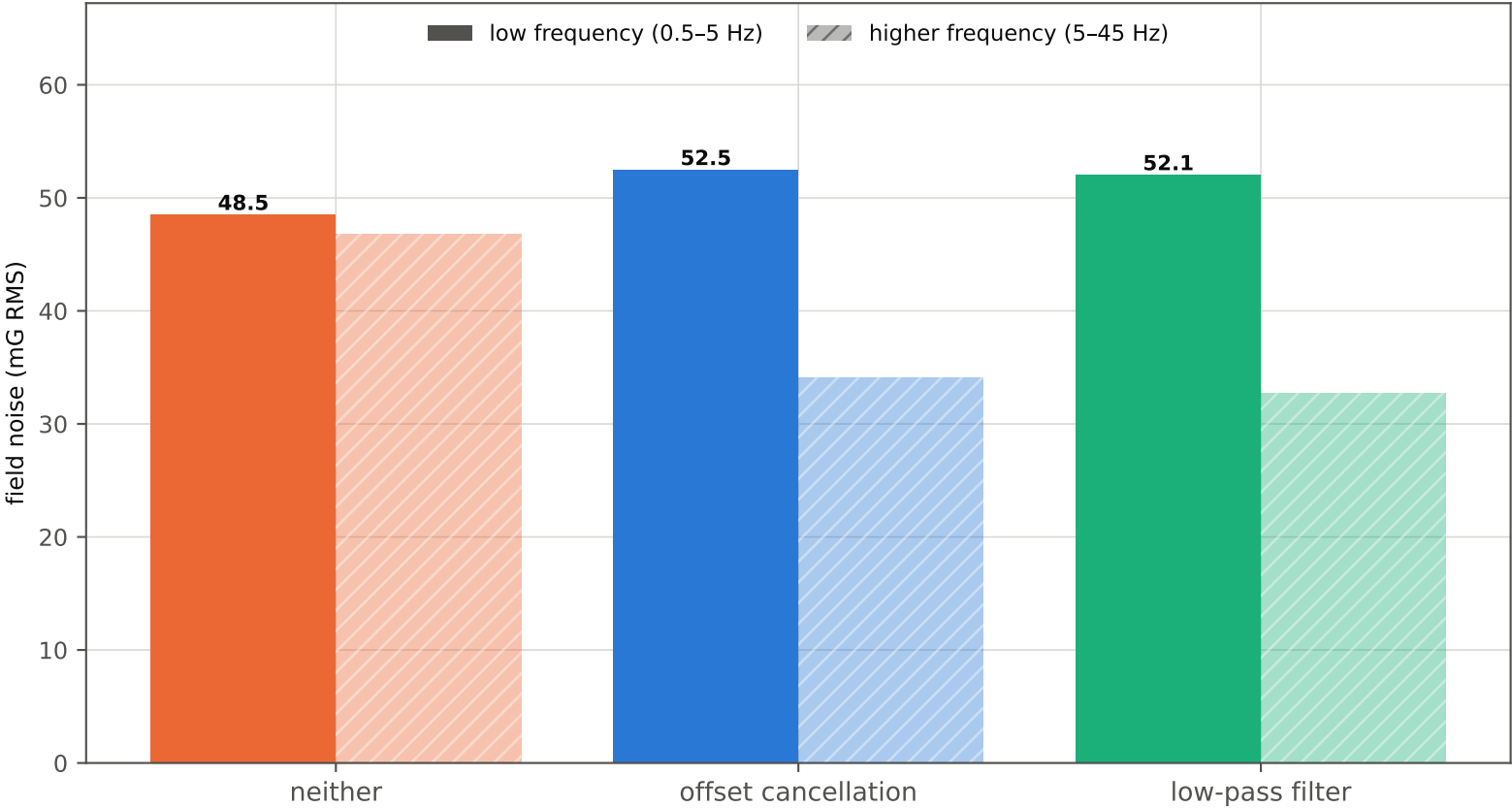
With offset cancellation off, the reading is shifted by a fixed amount— about 300 to 500 mG depending on the recording. The 'neither versus low-pass filter' comparison (the reference, gray / rightmost bars) stays near zero, showing the shift comes from turning off offset cancellation and not from the smoothing the low-pass filter shares. In the two rotating recordings the setting was identified from the data.

Noise in the reading, sensor held still



How much noise is in the reading at each frequency, with the sensor held still. Offset cancellation (blue) and the low-pass filter (green) both reduce the noise compared with neither (orange). Lower is quieter.

Noise in flight, motors running



Recorded while flying with the motors running. In the low-frequency band that matters for heading (solid bars), the noise is about the same for all three settings; the small spread is from the vehicle moving, not from the setting. The higher-frequency band (hatched) is lower with offset cancellation or the low-pass filter, from their smoothing.